

Image Processing and Analysis

Lecture 8、 Image Compression

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Outline

- 1 Basis of Image Compression
- 2 Image Compression Model
- 3 Elements of Information Theory
- 4 Error-Free Compression
- 5 Lossy Compression

Basis Concepts

- The problem solved by image compression is to minimize the amount of data required to represent a digital image. The basic principle of reducing data volume is to remove redundant data.
- The term "data compression" refers to reducing the amount of data required to represent a given amount of information. A clear distinction must be made between data and information.
- Data redundancy is the main issue in digital image compression. It is an entity that can be quantified mathematically.

Data Redundancy

- If n_1 and n_2 represent the number of information units carried in two data sets representing the same information, then the Relative Data Redundancy R_D for the first data set (represented by n_1) can be defined as

$$R_D = 1 - \frac{1}{C_R}$$
$$C_R = \frac{n_1}{n_2}$$

- C_R is referred to as the Compression Ratio. When $n_1 = n_2$, $C_R = 1$, $R_D = 0$, the first expression of information relative to the second data set contains no redundant data. When $n_2 \ll n_1$, it implies significant compression and high redundancy. When $n_2 \gg n_1$, it indicates that the amount of data contained in the second set greatly exceeds the data volume of the original expression.

Data Redundancy

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

Data Redundancy -- Coding Redundancy

- Here, once again, we assume that a discrete random variable r_k within the interval $[0,1]$ represents the grayscale level of an image and the probability of occurrence for each grayscale level is:

$$p_r(r_k) = \frac{n_k}{n}, k = 0, 1, \dots, L - 1$$

- If $l(r_k)$ represents the number of bits used to express each r_k , then the average number of bits required to represent each pixel is:

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

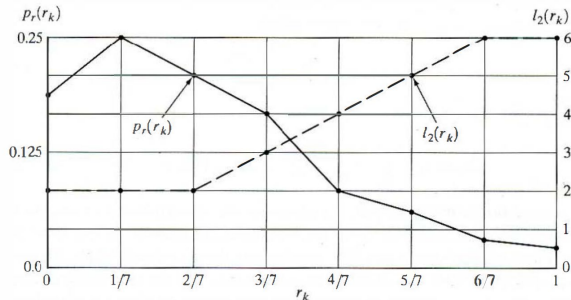
- The number of bits required to encode an image of size $M \times N$ is MNL_{avg}

Data Redundancy -- Coding Redundancy

r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_0 = 0$	0.19	000	3	11	2
$r_1 = 1/7$	0.25	001	3	01	2
$r_2 = 2/7$	0.21	010	3	10	2
$r_3 = 3/7$	0.16	011	3	001	3
$r_4 = 4/7$	0.08	100	3	0001	4
$r_5 = 5/7$	0.06	101	3	00001	5
$r_6 = 6/7$	0.03	110	3	000001	6
$r_7 = 1$	0.02	111	3	000000	6

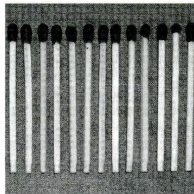
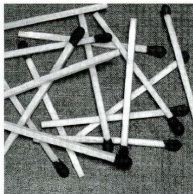
TABLE 8.1

Example of variable-length coding.

**FIGURE 8.1**

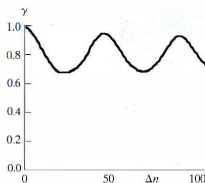
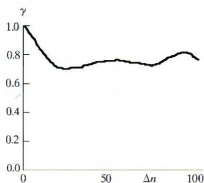
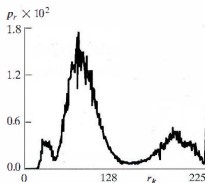
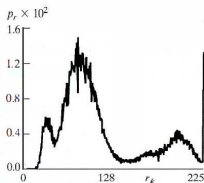
Graphic representation of the fundamental basis of data compression through variable-length coding.

Data Redundancy -- Interpixel Redundancy



a b
c d
e f

FIGURE 8.2 Two images and their gray-level histograms and normalized autocorrelation coefficients along one line.



Data Redundancy -- Interpixel Redundancy

- Figures 8.2(e) and (f) show the respective autocorrelation coefficients computed along one line of each image. These coefficients were computed using a normalized version of Eq. (4.6-30) in which

$$\gamma(\Delta n) = \frac{A(\Delta n)}{A(0)}$$

where

$$A(\Delta n) = \frac{1}{N - \Delta n} \sum_{y=0}^{N-1-\Delta n} f(x, y)f(x, y + \Delta n)$$

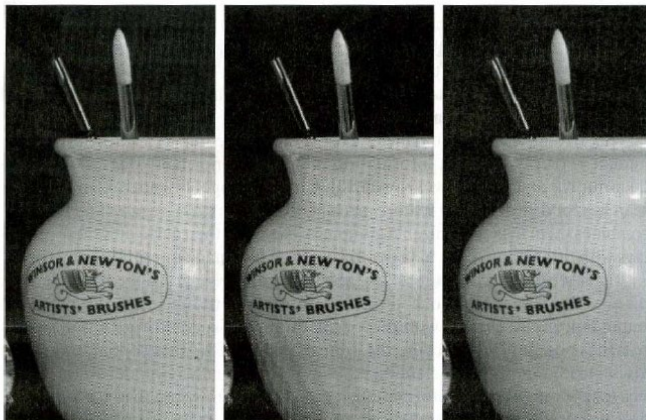
- A variety of names, including spatial redundancy, geometric redundancy, and interframe redundancy, have been coined to refer to these interpixel dependencies. We use the term interpixel redundancy to encompass them all.

Data Redundancy -- Psychovisual Redundancy

a b c

FIGURE 8.4

(a) Original image.

(b) Uniform quantization to 16 levels.
(c) IGS quantization to 16 levels.

Fidelity Criteria

- Removal of psychovisually redundant data results in a loss of real or quantitative visual information.
- Two general classes of criteria are used as the basis for such an assessment:
 - (1) objective fidelity criteria
 - (2) subjective fidelity criteria

Fidelity Criteria

- An example of objective fidelity criteria:

For any value of x and y , the error $e(x, y)$ between $f(x, y)$ and $\hat{f}(x, y)$ can be defined as

$$e(x, y) = \hat{f}(x, y) - f(x, y)$$

so that the total error between the two images is

$$\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]$$

The root-mean-square error e_{rms} between $f(x, y)$ and $\hat{f}(x, y)$ then is the square root of the squared error

$$e_{rms} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2 \right]^{1/2}$$

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Image Compression Model

- A compression system consists of two distinct structural blocks: an encoder and a decoder.
- In general, $\hat{f}(x, y)$ may or may not be an exact replica of $f(x, y)$. If it is, the system is error free or information preserving; if not, some level of distortion is present in the reconstructed image.
- Both the encoder and decoder shown in Fig. 8.5 consist of two relatively independent functions or subblocks: a source/channel encoder/decoder.

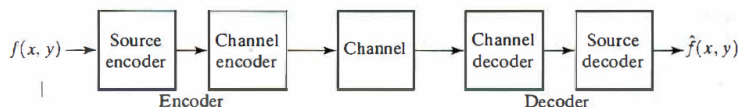
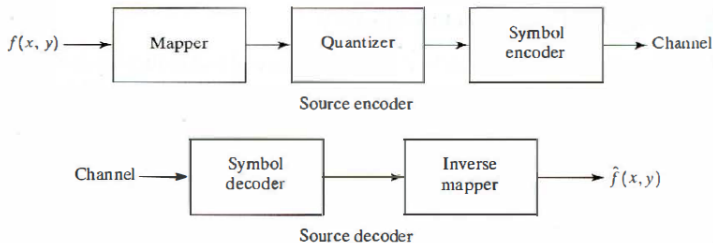


FIGURE 8.5 A general compression system model.

The Source Encoder and Decoder

- The source encoder is responsible for reducing or eliminating any coding, interpixel, or psychovisual redundancies in the input image.
- Normally, the source encoder can be modeled by a series of three independent operations: mapper, quantizer and symbol encoder.
- The source decoder contains only two components: symbol decoder and an inverse mapper.



The Channel Encoder and Decoder

- The channel encoder and decoder are designed to reduce the impact of channel noise by inserting a controlled form of redundancy into the source encoded data.
- As the output of the source encoder contains little redundancy, it would be highly sensitive to transmission noise without the addition of this "controlled redundancy."

The Channel Encoder and Decoder

- Hamming encoder and decoder

It is based on appending enough bits to the data being encoded to ensure that some minimum number of bits must change between valid code words.

- 7-bit Hamming (7,4) code word

$$h_1 = b_3 \oplus b_2 \oplus b_0 \quad h_3 = b_3$$

$$h_2 = b_3 \oplus b_1 \oplus b_0 \quad h_5 = b_2$$

$$h_4 = b_2 \oplus b_1 \oplus b_0 \quad h_6 = b_1$$

$$h_7 = b_0$$

- nonzero parity word

$$c_1 = h_1 \oplus h_3 \oplus h_5 \oplus h_7$$

$$c_2 = h_2 \oplus h_3 \oplus h_6 \oplus h_7$$

$$c_4 = h_4 \oplus h_5 \oplus h_6 \oplus h_7$$

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Measuring Information

- The fundamental premise of information theory is that the generation of information can be modeled as a probabilistic process that can be measured in a manner that agrees with intuition.
- A random event E that occurs with probability $P(E)$ is said to contain $I(E)$ units of information.

$$I(E) = \log \frac{1}{P(E)} = -\log P(E)$$

The quantity $I(E)$ often is called the self-information of E .

Measuring Information(CONT.)

- The base of the logarithm in Eq. (1) determines the unit used to measure information. If the base m logarithm is used, the measurement is said to be in m -ary units.

$$I(E) = \log \frac{1}{P(E)} = -\log P(E) \quad (1)$$

- If the base 2 is selected, the resulting unit of information is called a bit.

The Information Channel

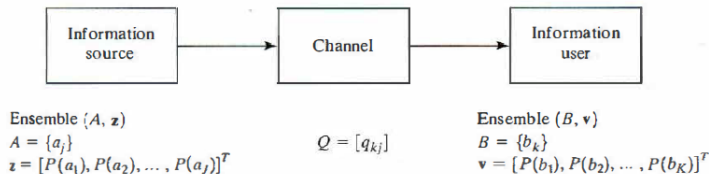


FIGURE 8.7 A simple information system.

- Assume that the information source in Fig generates a random sequence of symbols from a finite or countably infinite set of possible symbols. The set of source symbols $\{a_1, a_2, \dots, a_J\}$ is referred to as the source alphabet A , and the elements of the set are called symbols or letters.

The Information Channel(CONT.)

- The probability of the event that the source will produce symbol a_j is $P(a_j)$

$$\sum_{j=1}^J P(a_j) = 1$$

- A $J \times 1$ vector $z = [P(a_1), P(a_2), \dots, P(a_J)]^T$ customarily is used to represent the set of all source symbol probabilities $\{P(a_1), P(a_2), \dots, P(a_J)\}$. The finite ensemble (A, z) describes the information source completely.

The Information Channel(CONT.)

- The self-information generated by the production of a single source symbol is $I(a_j) = -\log P(a_j)$.
- If k source symbols are generated, for a sufficiently large value of k , symbol a_j will (on average) be output $kP(a_j)$ times. Thus the average self-information obtained from k outputs is

$$-kP(a_1) \log P(a_1) - kP(a_2) \log P(a_2) - \cdots - kP(a_j) \log P(a_j)$$

- or

$$-k \sum_{j=1}^J P(a_j) \log P(a_j)$$

The Information Channel(CONT.)

- The average information per source output, denoted $H(z)$, is

$$H(z) = - \sum_{j=1}^J P(a_j) \log P(a_j)$$

- This quantity is called the uncertainty or entropy of the source. It defines the average amount of information obtained by observing a single source output. As its magnitude increases, more uncertainty and thus more information is associated with the source. If the source symbols are equally probable, the entropy or uncertainty is maximized and the source provides the greatest possible average information per source symbol.

The Information Channel(CONT.)

- Because we modeled the input to the channel in *Fig.8.7* as a discrete random variable, the information transferred to the output of the channel is also a discrete random variable ($B : \{b_1, b_2, \dots, b_K\}$).
- The probability of the event that symbol b_k is presented to the information user is $P(b_k)$. The finite ensemble (B, v) , where $v = [P(b_1), P(b_2), \dots, P(b_K)]^T$, describes the channel output completely and thus the information received by the user.
- The probability $P(b_k)$ of a given channel output and the probability distribution of the source z are related by the expression

$$P(b_k) = \sum_{j=1}^J P(b_k|a_j)P(a_j)$$

The Information Channel(CONT.)

- $P(b_k|a_j)$ is the conditional probability that output symbol b_k is received, given that source symbol a_j was generated.
- The conditional probabilities are arranged in a matrix $K \times J$ matrix Q

$$Q = \begin{pmatrix} P(b_1|a_1) & P(b_1|a_2) & \cdots & P(b_1|a_J) \\ P(b_2|a_1) & P(b_2|a_2) & \cdots & P(b_2|a_J) \\ \vdots & \vdots & \ddots & \vdots \\ P(b_K|a_1) & P(b_K|a_2) & \cdots & P(b_K|a_J) \end{pmatrix}$$

- The probability distribution of the complete output alphabet can be computed from

$$v = Qz$$

- Matrix Q , with elements $q_{kj} = P(b_k|a_j)$, is referred to as the forward channel transition matrix or by the abbreviated term channel matrix.

The Information Channel(CONT.)

- Each b_k has one conditional entropy function, this conditional entropy function, denoted $H(z|b_k)$, can be written as

$$H(z|b_k) = - \sum_{j=1}^J P(a_j|b_k) \log P(a_j|b_k)$$

- $P(a_j|b_k)$ is the probability that symbol a_j was transmitted by the source, given that the user received b_k . The expected (average) value of this expression over all b_k is

$$H(z|v) = \sum_{k=1}^K H(z|b_k) P(b_k)$$

The Information Channel(CONT.)

- After substitution and some minor rearrangement $H(z|v)$ can be written as

$$H(z|v) = - \sum_{j=1}^J \sum_{k=1}^K P(a_j, b_k) \log P(a_j|b_k)$$

- $P(a_j, b_k)$ is the joint probability of a_j and b_k . That is, $P(a_j, b_k)$ is the probability that a_j is transmitted and b_k is received.
- The term $H(z|v)$ is called the equivocation of z with respect to v . It represents the average information of one source symbol, assuming observation of the output symbol that resulted from its generation.

The Information Channel(CONT.)

- Because $H(z)$ is the average information of one source symbol, assuming no knowledge of the resulting output symbol, the difference between $H(z)$ and $H(z|v)$ is the average information received upon observing a single output symbol. This difference, denoted $I(z, v)$ and called the mutual information of z and v , is

$$I(z, v) = H(z) - H(z|v)$$

- Substituting for $H(z)$ and $H(z|v)$, and recalling that $P(a_j) = P(a_j, b_1) + P(a_j, b_2) + \dots + P(a_j, b_K)$, yields

$$I(z, v) = \sum_{j=1}^J \sum_{k=1}^K P(a_j, b_k) \log \frac{P(a_j, b_k)}{P(a_j)P(b_k)}$$

- After further manipulation, can be written as

$$I(z, v) = \sum_{j=1}^J \sum_{k=1}^K P(a_j) q_{kj} \log \frac{q_{kj}}{\sum_{i=1}^J P(a_i) q_{ki}}$$

The Information Channel(CONT.)

- The average information received upon observing a single output of the information channel is a function of the input or source symbol probability vector z and channel matrix Q .
- The minimum possible value of $I(z, v)$ is zero and occurs when the input and output symbols are statistically independent, in which case $P(a_j, b_k) = P(a_j)P(b_k)$.
- The maximum value of $I(z, v)$ over all possible choices of source probabilities in vector z is the capacity, C , of the channel described by channel matrix Q . That is,

$$C = \max_z [I(z, v)]$$
- The capacity of the channel defines the maximum rate at which information can be transmitted reliably through the channel. Moreover, the capacity of a channel does not depend on the input probabilities of the source but is a function of the conditional probabilities defining the channel alone.

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Error-Free Compression

- In numerous applications error-free compression is the only acceptable means of data reduction. Its applications include the archival of medical or business documents, digital radiography, where lossy compression usually is prohibited for legal reasons.
- This section focuses on the classical, principal error-free compression strategies. They normally provide compression ratios of 2 to 10. Moreover, they are equally applicable to both binary and gray-scale images
- The error-free compression techniques generally are composed of two relatively independent operations: (1) devising an alternative representation of the image in which its interpixel redundancies are reduced; and (2) coding the representation to eliminate coding redundancies.

Error-Free Compression

- Variable-Length Coding
- LZW Coding
- Bit-Plane Coding
- Lossless Predictive Coding

Variable-Length Coding

- The simplest approach to error-free image compression is to reduce only coding redundancy
- Coding redundancy normally is present in any natural binary encoding of the gray levels in an image. As we noted in Section 8.1.1, it can be eliminated by coding the gray levels so that Eq. (8.1.4) is minimized.

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

- To do so requires construction of a variable-length code that assigns the shortest possible code words to the most probable gray levels.

Variable-Length Coding -- Huffman coding

- When coding the symbols of an information source individually, Huffman coding yields the smallest possible number of code symbols per-source symbol.
- In terms of the noiseless coding theorem (see Section 8.3.3), the resulting code is optimal for a fixed value of n , subject to the constraint that the source symbols be coded one at a time.

Original source		Source reduction				
Symbol	Probability	1	2	3	4	
a_2	0.4	0.4	0.4	0.4	0.6 0.4	
a_6	0.3	0.3	0.3	0.3		
a_1	0.1	0.1	0.2	0.3		
a_4	0.1	0.1				
a_3	0.06	0.1	0.1			
a_5	0.04					

FIGURE 8.11
Huffman source reductions.

Variable-Length Coding -- Huffman coding

- The first step is to create a series of source reductions by ordering the probabilities of the symbols under consideration and combining the lowest probability symbols into a single symbol that replaces them in the next source reduction.

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	0.4
a_1	0.1	0.1	0.2	0.3	
a_4	0.1	0.1	0.1		
a_3	0.06	0.1			
a_5	0.04				

FIGURE 8.11
Huffman source reductions.

Variable-Length Coding -- Huffman coding

- The second step is to code each reduced source, starting with the smallest source and working back to the original source. The minimal length binary code for a two-symbol source, of course, is the symbols 0 and 1.

$$\begin{aligned}
 L_{avg} &= (0.4)(1) + (0.3)(2) + (0.1)(3) + (0.1)(4) \\
 &\quad + (0.06)(5) + (0.04)(5) \\
 &= 2.2 \text{ bits/symbol}
 \end{aligned}$$

FIGURE 8.12
Huffman code
assignment
procedure.

Original source			Source reduction			
Sym.	Prob.	Code	1	2	3	4
a_2	0.4	1	0.4 1	0.4 1	0.4 1	0.6 0
a_6	0.3	00	0.3 00	0.3 00	0.3 00	0.4 1
a_1	0.1	011	0.1 011	0.2 010	0.3 01	
a_4	0.1	0100	0.1 0100	0.1 011		
a_3	0.06	01010	0.1 0101			
a_5	0.04	01011				

Variable-Length Coding -- Other near optimal variable length codes

Source symbol	Probability	Binary Code	Huffman	Truncated Huffman	B ₂ -Code	Binary Shift	Huffman Shift
<i>Block 1</i>							
a_1	0.2	00000	10	11	C00	000	10
a_2	0.1	00001	110	011	C01	001	11
a_3	0.1	00010	111	0000	C10	010	110
a_4	0.06	00011	0101	0101	C11	011	100
a_5	0.05	00100	00000	00010	C00C00	100	101
a_6	0.05	00101	00001	00011	C00C01	101	1110
a_7	0.05	00110	00010	00100	C00C10	110	1111
<i>Block 2</i>							
a_8	0.04	00111	00011	00101	C00C11	111000	0010
a_9	0.04	01000	00110	00110	C01C00	111001	0011
a_{10}	0.04	01001	00111	00111	C01C01	111010	00110
a_{11}	0.04	01010	00100	01000	C01C10	111011	00100
a_{12}	0.03	01011	01001	01001	C01C11	111100	00101
a_{13}	0.03	01100	01110	100000	C10C00	111101	001110
a_{14}	0.03	01101	01111	100001	C10C01	111110	001111
<i>Block 3</i>							
a_{15}	0.03	01110	01100	100010	C10C10	111111000	000010
a_{16}	0.02	01111	010000	100011	C10C11	111111001	000011
a_{17}	0.02	10000	010001	100100	C11C00	111111010	0000110
a_{18}	0.02	10001	001010	100101	C11C01	111111011	0000100
a_{19}	0.02	10010	001011	100110	C11C10	111111100	0000101
a_{20}	0.02	10011	011010	100111	C11C11	111111101	00001110
a_{21}	0.01	10100	011011	101000	C00C00C00	111111110	00001111
<i>Entropy</i> 4.0							
<i>Average length</i>							
		5.0	4.05	4.24	4.65	4.59	4.13

TABLE 8.5
Variable-length codes

Variable-Length Coding -- Arithmetic Coding

- Unlike the variable-length codes described previously, arithmetic coding generates nonblock codes. It achieves (but only in theory) the bound established by the noiseless coding theorem.

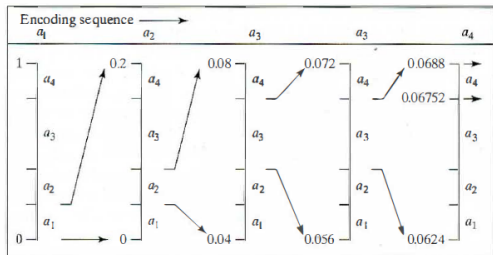


FIGURE 8.13
Arithmetic coding procedure.

Source Symbol	Probability	Initial Subinterval
a_1	0.2	[0.0, 0.2)
a_2	0.2	[0.2, 0.4)
a_3	0.4	[0.4, 0.8)
a_4	0.2	[0.8, 1.0)

TABLE 8.6
Arithmetic coding example.

LZW Coding

- Having examined the principal methods for removing coding redundancy, we now consider one of several error-free compression techniques that also attack an image's interpixel redundancies.
- LZW coding assigns fixed-length code words to variable length sequences of source symbols but requires no a priori knowledge of the probability of occurrence of the symbols to be encoded.

$$\begin{bmatrix} 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \\ 39 & 39 & 126 & 126 \end{bmatrix}$$

LZW Coding

- At the onset of the Coding process, a code book or "dictionary" containing the source symbols to be coded is constructed. For 8-bit monochrome images, the first 256 words of the dictionary are assigned to the gray values 0, 1, 2, . . . , 255.

Dictionary Location	Entry
0	0
1	1
⋮	⋮
255	255
256	—
⋮	⋮
511	—

LZW Coding

Currently Recognized Sequence	Pixel Being Processed	Encoded Output	Dictionary Location (Code Word)	Dictionary Entry
	39			
39	39	39	256	39-39
39	126	39	257	39-126
126	126	126	258	126-126
126	39	126	259	126-39
39	39			
39-39	126	256	260	39-39-126
126	126			
126-126	39	258	261	126-126-39
39	39			
39-39	126			
39-39-126	126	260	262	39-39-126-126
126	39			
126-39	39	259	263	126-39-39
39	126			
39-126	126	257	264	39-126-126
126		126		

TABLE 8.7
LZW coding
example.

Bit-Plane Coding

- Another effective technique for reducing an image's interpixel redundancies is to process the image's bit planes individually.
- The technique, called bit-plane coding, is based on the concept of decomposing a multilevel (monochrome or color) image into a series of binary images and compressing each binary image via one of several well-known binary compression methods.
- In this section, we describe the most popular decomposition approaches and review several of the more commonly used compression methods.

Bit-Plane Coding

- Bit-plane decomposition

The gray levels of an m -bit gray-scale image can be represented in the form of the base 2 polynomial:

$$a_{m-1}2^{m-1} + a_{m-2}2^{m-2} + \cdots + a_1 2^1 + a_0 2^0$$

Based on this property, a simple method of decomposing the image into a collection of binary images is to separate the m coefficients of the polynomial into m 1-bit *bit planes*.

An alternative decomposition approach (which reduces the effect of small gray-level variations) is to first represent the image by an m -bit *Gray code*.

$$g_i = a_i \oplus a_{i+1}, \quad 0 \leq i \leq m-2$$

$$g_{m-1} = a_{m-1}$$

Bit-Plane Coding

- Bit-plane decomposition



his Indenture made this run
 his year of our Lord one thou
 indrinity six between Stockley
 f Kny And Stat of Tennessee
 Andrew Jackson of the counte
 stat of said of the other part
 said Stockley Donelson for A
 of the Sum of two thousand
 hand paid the receipt where
 hath And by these presents
 self alien encoff And confer
 Jackson his theirs And a
 certain traits or parole of La
 sand aces / one thousand are
 and a ball line and his

a b

FIGURE 8.14 A

1024 × 1024

(a) 8-bit
 monochrome
 image and
 (b) binary image.

Bit-Plane Coding

- Bit-plane decomposition



Bit 7



Bit 7



Bit 6



Bit 6



Bit 5



Bit 5



Bit 4

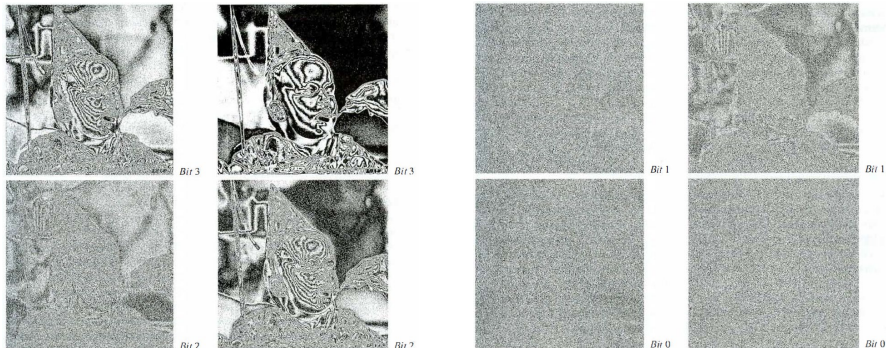


Bit 4

The four most significant binary (left column) and Gray-coded (right column) bit planes.

Bit-Plane Coding

- Bit-plane decomposition



The four least significant binary (left column) and Gray-coded (right column) bit planes.

Bit-Plane Coding

- Constant area coding

A simple but effective method of compressing a binary image or bit plane is to use special code words to identify large areas of contiguous 1's or 0's.

In one such approach, called *constant area coding* (CAC), the image is divided into blocks of size $p \times q$ pixels, which are classified as all white, all black, or mixed intensity.

The most probable or frequently occurring category is then assigned the 1-bit code word 0, and the other two categories are assigned the 2-bit codes 10 and 11. Compression is achieved because the pq bits that normally would be used to represent each constant area are replaced by a 1-bit or 2-bit code word.

Bit-Plane Coding

- One/Two-dimensional run-length coding

An effective alternative to constant area coding is to represent each row of an image or bit plane by a sequence of lengths that describe successive runs of black and white pixels.

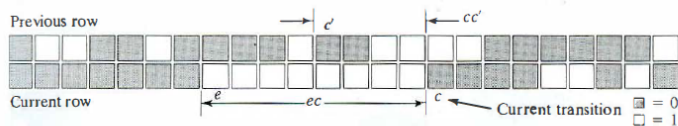


FIGURE 8.17 A relative address coding (RAC) illustration.

Distance measured	Distance	Code	Distance range	Code $h(d)$
cc'	0	0	1 – 4	0xx
ec or cc' (left)	1	100	5 – 20	10xxxx
cc' (right)	1	101	21 – 84	110xxxxxx
ec	$d (d > 1)$	111 $h(d)$	85 – 340	1110xxxxxxxxxx
cc' (c' to left)	$d (d > 1)$	1100 $h(d)$	341 – 364	11110xxxxxxxxxx
cc' (c' to right)	$d (d > 1)$	1101 $h(d)$	1365 – 5460	111110xxxxxxxxxx

Bit-Plane Coding

• Contour tracing and coding

- Relative address coding is to represent intensity transitions that make up the contours in a binary image.
- Another approach is to represent each contour by a set of boundary points or by a single boundary point and a set of directionals.
- In this section, we describe *predictive differential quantizing* (PDQ), which demonstrates the essential characteristics of both approaches (scan-line-oriented contour tracking).

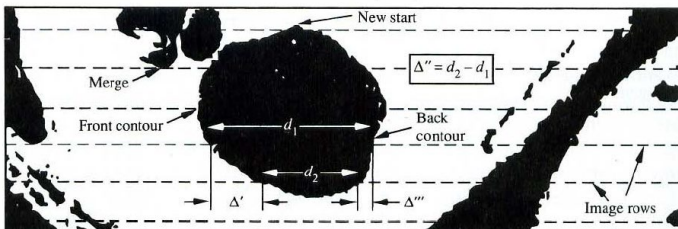


FIGURE 8.18 Parameters of the PDQ algorithm.

Bit-Plane Coding

	Bit-plane code rate (bits/pixel)								Code Rate	Compression Ratio
Method	7	6	5	4	3	2	1	0		
<i>Binary Bit-Plane Coding</i>										
CBC (4 × 4)	0.14	0.24	0.60	0.79	0.99	—	—	—	5.75	1.4:1
RLC	0.09	0.19	0.51	0.68	0.87	1.00	1.00	1.00	5.33	1.5:1
PDQ	0.07	0.18	0.79	—	—	—	—	—	6.04	1.3:1
DDC	0.07	0.18	0.79	—	—	—	—	—	6.03	1.3:1
RAC	0.06	0.15	0.62	0.91	—	—	—	—	5.17	1.4:1
<i>Gray Bit-Plane Coding</i>										
CBC (4 × 4)	0.14	0.18	0.48	0.40	0.61	0.98	—	—	4.80	1.7:1
RLC	0.09	0.13	0.40	0.33	0.51	0.85	1.00	1.00	4.29	1.9:1
PDQ	0.07	0.12	0.61	0.40	0.82	—	—	—	5.02	1.6:1
DDC	0.07	0.11	0.61	0.40	0.81	—	—	—	5.00	1.6:1
RAC	0.06	0.10	0.49	0.31	0.62	—	—	—	4.05	1.8:1

TABLE 8.8

Error-free bit-plane coding results for Fig. 8.14(a):
 $H \approx 6.82$ bits/pixel

	WBS (1×8)	WBS (4×4)	RLC	PDQ	DDC	RAC
<i>Code rate (bits/pixel)</i>	0.48	0.39	0.32	0.23	0.22	0.23
<i>Compression ratio</i>	2.1:1	2.6:1	3.1:1	4.4:1	4.7:1	4.4:1

TABLE 8.9

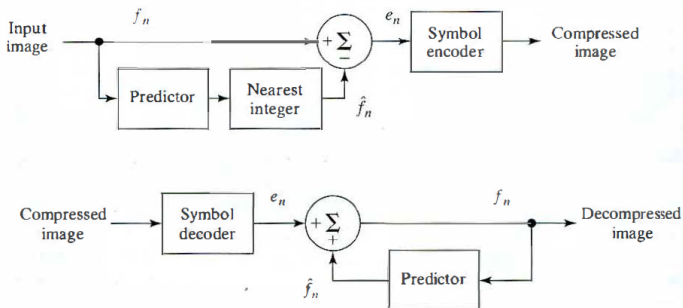
Error-free binary image compression results for Fig. 8.14(b):
 $H \approx 0.55$ bits/pixel.

Lossless Predictive Coding

- Lossless Predictive Coding is based on eliminating the interpixel redundancies of closely spaced pixels by extracting and coding only the new information in each pixel. The new information of a pixel is defined as the difference between the actual and predicted value of that pixel.

a
b

FIGURE 8.19 A
lossless predictive
coding model:
(a) encoder;
(b) decoder.



Lossless Predictive Coding

- Prediction error is coded using a variable-length code.

$$e_n = f_n - \hat{f}_n$$

- The decoder reconstructs e_n from the received variable-length code words and performs the inverse operation

$$f_n = e_n + \hat{f}_n$$

- In most cases, however, the prediction is formed by a linear combination of m previous pixels. That is,

$$\hat{f}_n = \text{round} \left[\sum_{i=1}^m \alpha_i f_{n-i} \right]$$

- In other cases, n is used as an index on the spatial coordinates and/or frame number (in a time sequence of images) of an image

$$\hat{f}_n(x, y) = \text{round} \left[\sum_{i=1}^m \alpha_i f(x, y - i) \right]$$

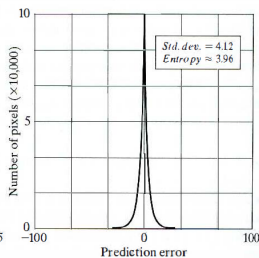
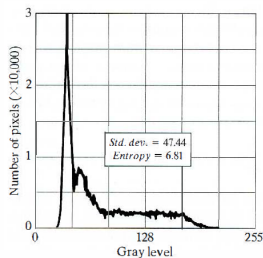
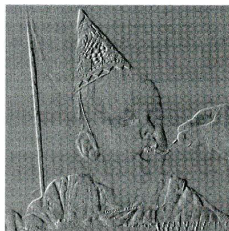
Lossless Predictive Coding

$$\hat{f}(x, y) = \text{round}[\alpha f(x, y - 1)]$$

a
b c

FIGURE 8.20

(a) The prediction error image resulting from Eq. (8.4-9).
(b) Gray-level histogram of the original image.
(c) Histogram of the prediction error.



Outline

- 1 Basis of Image Compression
- 2 Image Compression Model
- 3 Elements of Information Theory
- 4 Error-Free Compression
- 5 Lossy Compression

Lossy Compression

- Unlike the error-free approaches outlined in the previous section, lossy encoding is based on the concept of compromising the accuracy of the reconstructed image in exchange for increased compression.
- If the resulting distortion (which may or may not be visually apparent) can be tolerated, the increase in compression can be significant.
- Many lossy encoding techniques are capable of reproducing recognizable monochrome images from data that have been compressed by more than 100:1 and images that are virtually indistinguishable from the originals at 10:1 to 50:1. Error-free encoding of monochrome images, however, seldom results in more than a 3 : 1 reduction in data.

Lossy Compression

- Lossy Predictive Coding
- Transform Coding
- Wavelet Coding

Lossy Compression -- Lossy Predictive Coding

- Adding a quantizer to the model introduced in Section 8.4.4 and examining the resulting trade-off between reconstruction accuracy and compression performance.

$$\hat{f}_n = \hat{e}_n + \hat{f}_n$$

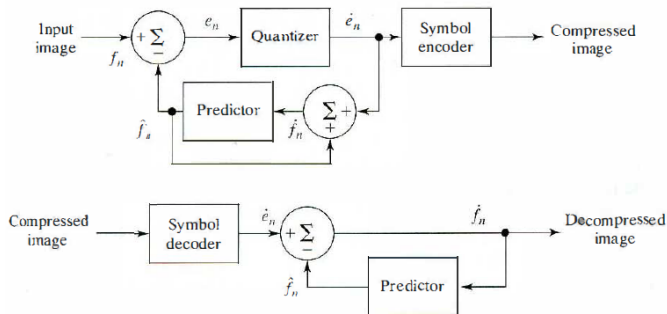
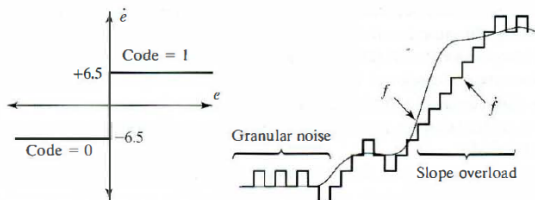


FIGURE 8.21 A lossy predictive coding model: (a) encoder and (b) decoder.

Lossy Compression -- Lossy Predictive Coding



a b
c

FIGURE 8.22 An example of delta modulation.

Input		Encoder				Decoder		Error
n	f	$\hat{f}$	e	$\hat{e}$	$\hat{f}$	$\hat{f}$	$\hat{f}$	$[f - \hat{f}]$
0	14	—	—	—	14.0	—	14.0	0.0
1	15	14.0	1.0	6.5	20.5	14.0	20.5	-5.5
2	14	20.5	-6.5	-6.5	14.0	20.5	14.0	0.0
3	15	14.0	1.0	6.5	20.5	14.0	20.5	-5.5
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
14	29	20.5	8.5	6.5	27.0	20.5	27.0	2.0
15	37	27.0	10.0	6.5	33.5	27.0	33.5	3.5
16	47	33.5	13.5	6.5	40.0	33.5	40.0	7.0
17	62	40.0	22.0	6.5	46.5	40.0	46.5	15.5
18	75	46.5	28.5	6.5	53.0	46.5	53.0	22.0
19	77	53.0	24.0	6.5	59.6	53.0	59.6	17.5
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Lossy Compression -- Lossy Predictive Coding

- Optimal predictors

The optimal predictor used in most predictive coding applications minimizes the encoder's mean-square prediction error

$$E\{e_n^2\} = E\{[f_n - \hat{f}_n]^2\}$$

subject to the constraint that

$$\dot{f}_n = \dot{e}_n + \hat{f}_n \approx e_n + \hat{f}_n = f_n$$

and

$$\hat{f}_n = \sum_{i=1}^m \alpha_i f_{n-i}$$

Lossy Compression -- Lossy Predictive Coding

- Optimal predictors

Under these conditions, the optimal predictor design problem is reduced to the relatively straightforward exercise of selecting the m prediction coefficients that minimize the expression

$$E\{e_n^2\} = E\left\{\left[f_n - \sum_{i=1}^m \alpha_i f_{n-i}\right]^2\right\}$$

Differentiating the above Eq. with respect to each coefficient, equating the derivatives to zero, and solving the resulting set of simultaneous equations under the assumption that f_n has mean zero and variance σ^2 yields

$$\alpha = R^{-1}r$$

Lossy Compression -- Lossy Predictive Coding

- Optimal predictors

$$\alpha = R^{-1}r$$

where R^{-1} is the inverse of the $m \times m$ autocorrelation matrix

$$R = \begin{bmatrix} E\{f_{n-1}f_{n-1}\} & \cdots & E\{f_{n-1}f_{n-m}\} \\ \vdots & \ddots & \vdots \\ E\{f_{n-m}f_{n-1}\} & \cdots & E\{f_{n-m}f_{n-m}\} \end{bmatrix}$$

and r and α are the m -element vectors

$$r = \begin{bmatrix} E\{f_n f_{n-1}\} \\ E\{f_n f_{n-2}\} \\ \vdots \\ E\{f_n f_{n-m}\} \end{bmatrix} \quad \alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{bmatrix}$$

Thus for any input image, the coefficients that minimize Eq. (8.5-7) can be determined via a series of elementary matrix operations.

Lossy Compression -- Lossy Predictive Coding

- Optimal predictors

The variance of the prediction error that results from the use of these optimal coefficients is

$$\sigma_e^2 = \sigma^2 - \boldsymbol{\alpha}^T \mathbf{r} = \sigma^2 - \sum_{i=1}^m E\{f_n f_{n-i}\} \alpha_i$$

The sum of the prediction coefficients α normally is required to be less than or equal to one as

$$\sum_{i=1}^m \alpha_i \leq 1$$

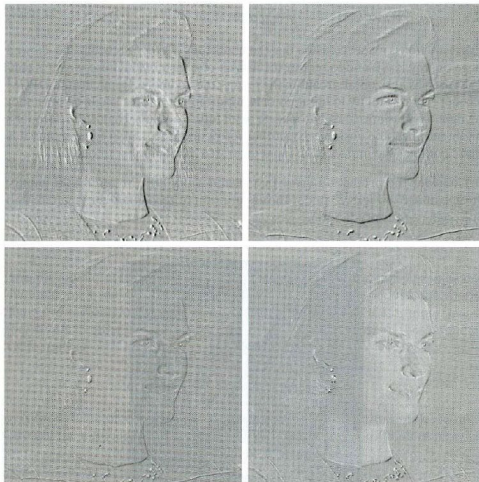
This restriction is made to ensure that the predictor's output falls within the allowed range of gray levels and to reduce the impact of transmission noise, which is generally seen as horizontal streaks in the reconstructed image.

Lossy Compression -- Lossy Predictive Coding

- Optimal predictors

a b
c d

FIGURE 8.24 A comparison of four linear prediction techniques.



Lossy Compression -- Lossy Predictive Coding

- Optimal quantization

$$t = q(s)$$

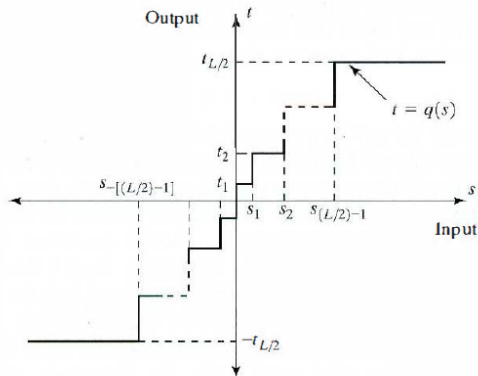


FIGURE 8.25 A typical quantization function.

Lossy Compression -- Lossy Predictive Coding

- Optimal quantization

The quantizer design problem is to select the best s_i and t_i for a particular optimization criterion and input probability density function $p(s)$.

If the optimization criterion, which could be either a statistical or psychovisual measure is the minimization of the mean-square quantization error and $p(s)$ is an even function, the conditions for minimal error are

$$\int_{s_{i-1}}^{s_i} (s - t)p(s)ds = 0, i = 1, 2, \dots, \frac{L}{2}$$

$$s_i = \begin{cases} 0 & i = 0 \\ \frac{t_i + t_{i+1}}{2} & i = 1, 2, \dots, \frac{L}{2} - 1 \\ \infty & i = L/2 \end{cases} \quad \begin{aligned} s_{-i} &= -s_i \\ t_{-i} &= -t_i \end{aligned}$$

Lossy Compression -- Lossy Predictive Coding

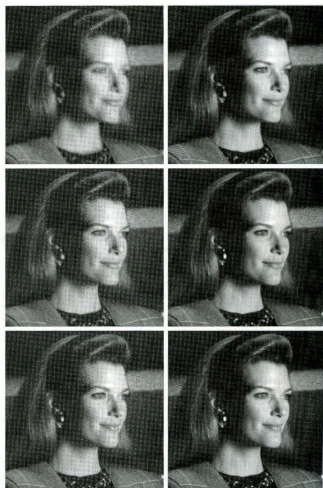
- Optimal quantization

TABLE 8.10
Lloyd-Max
quantizers for a
Laplacian
probability
density function
of unit variance.

Levels i	2		4		8	
	s_i	t_i	s_i	t_i	s_i	t_i
1	∞	0.707	1.102	0.395	0.504	0.222
2			∞	1.810	1.181	0.785
3					2.285	1.576
4					∞	2.994
θ	1.414		1.087		0.731	

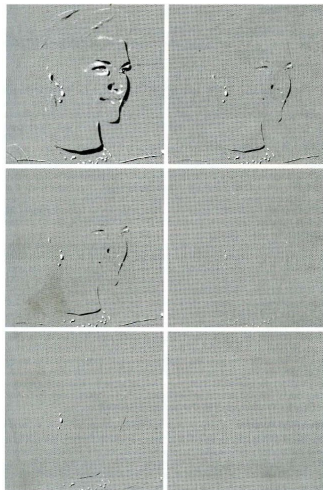
$$t_i - t_{i-1} = s_i - s_{i-1} = \theta$$

Lossy Compression -- Lossy Predictive Coding



a b
c d
e f

FIGURE 8.26 DPCM result images: (a) 1.0; (b) 1.125; (c) 2.0; (d) 2.125; (e) 3.0; (f) 3.125 bits/pixel.



a b
c d
e f

FIGURE 8.27 The scaled ($\times 8$) DPCM error images that correspond to Figs. 8.26(a) through (f).

Lossy Compression -- Transform Coding

- The predictive coding techniques operate directly on the pixels of an image and thus are spatial domain methods.
- Transform Coding considers compression techniques that are based on modifying the transform of an image.
- In transform coding, a reversible, linear transform (such as the Fourier transform) is used to map the image into a set of transform coefficients, which are then quantized and coded.

Lossy Compression -- Transform Coding

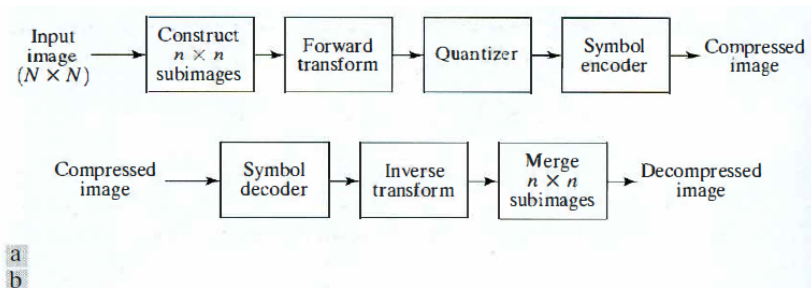


FIGURE 8.28 A transform coding system: (a) encoder; (b) decoder.

Lossy Compression -- Transform Coding

- Transform selection

Consider an image $f(x, y)$ of size $N \times N$ whose forward, discrete transform, $T(u, v)$, can be expressed in terms of the general relation

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) g(x, y, u, v)$$

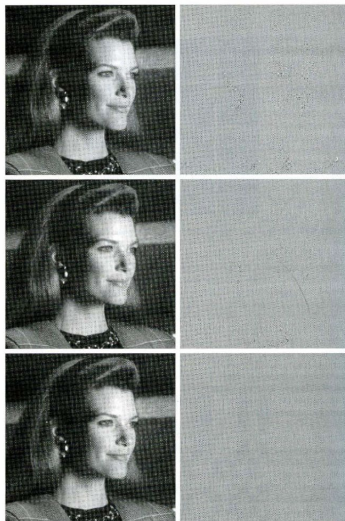
for $u, v = 0, 1, 2, \dots, N - 1$.

Given $T(u, v)$, $f(x, y)$ similarly can be obtained using the generalized inverse discrete transform

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) h(x, y, u, v)$$

Lossy Compression -- Transform Coding

- Transform selection



a b
c d
e f

FIGURE 8.31 Approximations of Fig. 8.23 using the (a) Fourier, (c) Hadamard, and (e) cosine transforms, together with the corresponding scaled error images.

Lossy Compression -- Transform Coding

- Subimage size selection

Another significant factor affecting transform coding error and computational complexity is subimage size.

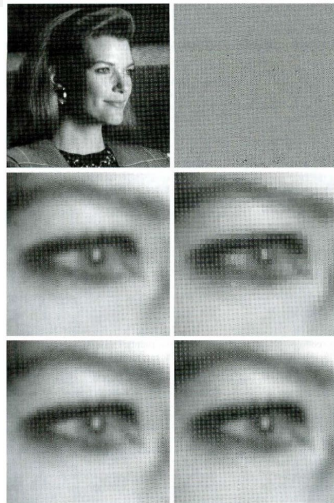
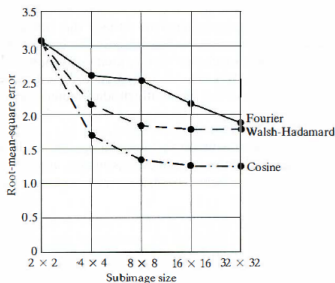
In most applications, images are subdivided so that the correlation (redundancy) between adjacent subimages is reduced to some acceptable level and so that n is an integer power of 2 where n is the subimage dimension.

In general, both the level of compression and computational complexity increase as the subimage size increases. The most popular subimage sizes are 8 X 8 and 16 X 16 .

Lossy Compression -- Transform Coding

- Subimage size selection

FIGURE 8.33
Reconstruction
error versus
subimage size.



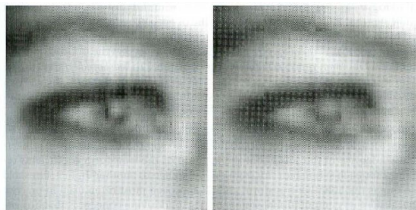
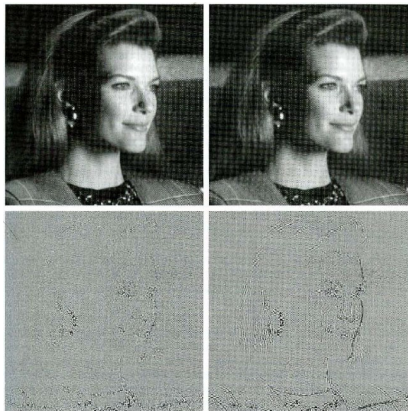
a b
c d
e f

FIGURE 8.34 Approximations of Fig. 8.23 using 25% of the DCT coefficients: (a) and (b) 8×8 subimage results; (c) zoomed original; (d) 2×2 result; (e) 4×4 result; and (f) 8×8 result.

Lossy Compression -- Transform Coding

- Bit allocation

The overall process of truncating, quantizing, and coding the coefficients of a transformed subimage is commonly called bit allocation.



a
b
c
d
e
f

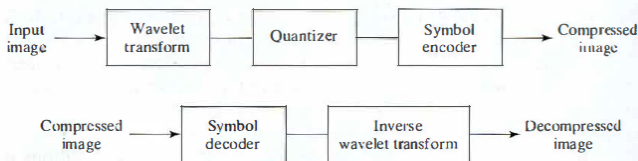
FIGURE 8.35 Approximations of Fig. 8.23 using 12.5% of the 8×8 DCT coefficients: (a), (c), and (e) threshold coding results; (b), (d), and (f) zonal coding results.

Lossy Compression -- Wavelet Coding

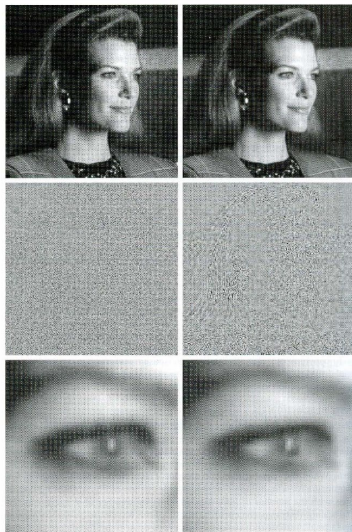
- Like the transform coding techniques of the previous section, wavelet coding is based on the idea that the coefficients of a transform that decorrelates the pixels of an image can be coded more efficiently than the original pixels themselves.

a
b

FIGURE 8.39 A wavelet coding system:
(a) encoder;
(b) decoder.

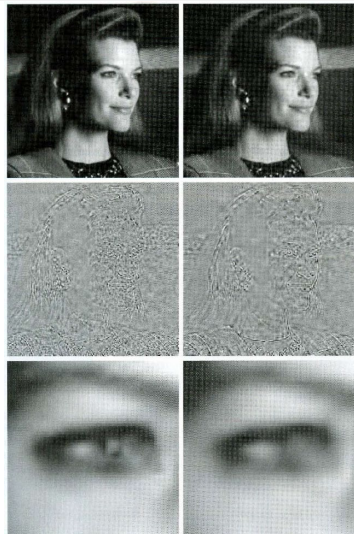


Lossy Compression -- Wavelet Coding



a b
c d
e f

FIGURE 8.40 (a), (c), and (e) Wavelet coding results comparable to the transform-based results in Figs. 8.38(a), (c), and (e); (b), (d), and (f) similar results for Figs. 8.38(b), (d), and (f).



a b
c d
e f

FIGURE 8.41 (a), (c), and (e) Wavelet coding results with a compression ratio of 108 to 1; (b), (d), and (f) similar results for a compression of 167 to 1.

Lossy Compression -- Wavelet Coding

- Wavelet selection

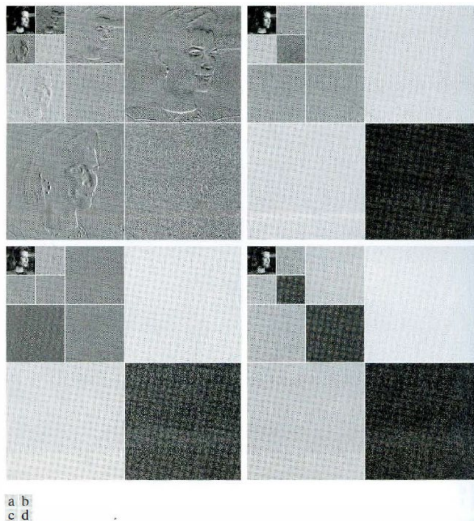


FIGURE 8.42 Wavelet transforms of Fig. 8.23 with respect to (a) Haar wavelets, (b) Daubechies wavelets, (c) symlets, and (d) Cohen-Daubechies-Feauveau biorthogonal wavelets.

Lossy Compression -- Wavelet Coding

- Decomposition level selection

Another factor affecting wavelet coding computational complexity and reconstruction error is the number of transform decomposition levels.

Scales and Filter Bank Iterations	Approximation Coefficient Image	Truncated Coefficients (%)	Reconstruction Error (rms)
1	256×256	75%	1.93
2	128×128	93%	2.69
3	64×64	97%	3.12
4	32×32	98%	3.25
5	16×16	98%	3.27

TABLE 8.13

Decomposition level impact on wavelet coding the 512×512 image of Fig. 8.23.

Lossy Compression -- Wavelet Coding

- Quantizer Designing

The major factor influencing the compression and reconstruction errors in wavelet coding is the quantization of coefficients. Although the most widely applied quantizer is uniform, the effectiveness of quantization can be further improved by:

- (1) introducing an enlarged quantization interval centered at zero, which is known as the dead zone, or
- (2) adaptively adjusting the size of the quantization intervals from one scale to another.